

CONCLUSION

The study on EEG-based classification for early detection of Major Depressive Disorder (MDD) highlights the effectiveness of machine learning techniques in analysing neural activity and identifying patterns associated with mental health conditions. The project involved a systematic workflow, starting from EEG data collection and preprocessing to feature extraction, model selection, and classification. Each step played a crucial role in enhancing the accuracy and reliability of the final predictions.

Preprocessing proved to be a vital step in improving signal quality by removing artifacts, noise, and unwanted interference. The application of filters such as notch filtering, bandpass filtering, and Independent Component Analysis (ICA) helped in preserving meaningful neural activity while eliminating external disturbances. This ensured that the extracted features were representative of actual brain dynamics rather than artifacts introduced during data collection.

Feature extraction focused on identifying key attributes of the EEG signals that contribute to distinguishing different mental states. By utilizing time-domain, frequency-domain, and time-frequency features, the extracted parameters provided a comprehensive representation of neural activity. Frequency-specific features, such as power in delta, theta, alpha, beta, and gamma bands, played a crucial role in understanding cognitive states, while statistical measures such as variance, skewness, and kurtosis contributed to signal characterization.

Model selection and classification involved evaluating multiple approaches to identify the most effective learning framework for EEG-based MDD detection. The choice of model depended on factors such as feature complexity, dataset size, and classification accuracy. To ensure robust generalization, techniques like cross-validation and hyperparameter tuning were applied, optimizing the model's ability to distinguish between healthy individuals and those with MDD.

The results demonstrated that machine learning models can successfully classify EEG signals related to depressive symptoms, reinforcing the potential of EEG-based biomarkers in mental health diagnostics. However, challenges remain, including inter-subject variability, dataset limitations, and the need for real-time adaptability in practical applications. Future advancements may focus on improving model interpretability, integrating deep learning.

architectures for more complex feature representations and developing personalized diagnostic systems that adapt to individual brain patterns.